

Leo Ng

M: +61 452 385 582

E: leongheileong@gmail.com

L: linkedin.com/in/hei-leong-leo-ng

Profile

- Bachelor of Engineering (Electrical) and Computer Science – penultimate year at UNSW with a Distinction average
- Hands-on experience with embedded sensing systems, microcontrollers, PCB design, sensor integration and hardware troubleshooting
- Developed and tested RP2040-based electronics using ToF sensors, I²C peripherals, power regulation and custom PCB hardware

Education

Bachelor of Engineering (Electrical) and Computer Science **2023 – Expected Aug 2027**
UNSW Sydney

- Achieved a Distinction average

Higher School Certificate **2022**
Epping Boys High School

- ATAR: 98.70

Relevant Experience

Embedded Sensing System / Piano Teaching Aid **May 2026 – Aug 2026**
(UNSW Course Engineering Project)

- Designed an RP2040-based sensing system using 13 VL53L1X time-of-flight sensors for real-time key and hand-position detection
- Integrated multiple I²C sensors using XSHUT control and address assignment, and debugged communication timeouts, startup conflicts and inconsistent readings
- Designed PCB schematics and layouts in KiCad, including sensor interfaces, power regulation, connectors, LEDs and microcontroller circuitry
- Performed hands-on testing and troubleshooting of power, signal and sensor behaviour, including battery-related LED instability, audio noise and reflective-surface sensing issues

Technologies: RP2040, C/MicroPython, VL53L1X, I²C, KiCad, PCB Design, Embedded Systems, Hardware Debugging

Custom Flip Phone – Hardware & Software Integration **Aug 2026 – Present**
(Personal Engineering Project)

- Developing a custom compact flip-phone platform integrating an Orange Pi 3B, 4.97-inch MIPI DSI display, keypad controls and Android-based software

- Investigated display controller pinouts, MIPI DSI lane configuration, voltage requirements and Linux/Android panel-driver integration for hardware bring-up
- Built a Kotlin/Jetpack Compose launcher with physical-key navigation, linking software behaviour to the constraints of the custom hardware interface

Technologies: Orange Pi, MIPI DSI, Android, Kotlin, Linux, Hardware/Software Integration

Employment

Restaurant Team Member & Operations Support

Pho Sam Lane Cove

Dec 2022 – Current

- Assisted with operational responsibilities including stocktake, inventory tracking, invoicing and day-to-day workflow support
- Provided technical support and troubleshooting for business systems, devices and technology-related issues within the restaurant

Technical Skills

- **Programming:** C, Python, JavaScript, Shell, Kotlin
- **Embedded & Hardware:** RP2040, I²C, sensor integration, PCB design, power regulation, hardware/firmware troubleshooting
- **Engineering Software & Tools:** KiCad, MATLAB, Simulink, LTspice, Git, Linux
- **Testing & Integration:** circuit testing, debugging, component selection, hardware validation, hardware/software integration

Languages

- English – Fluent
- Cantonese – Native
- Mandarin – Conversational
- Japanese – Basic
- German – Basic (A2 Level)